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CO5-AT3 - Mock Interview

- 1) Suppose you are working on an industrial monitoring system. How would you use Bayesian networks to predict machine failures?
 - A) A Bayesian network represents relationships between conditions like temperature, pressure and failure. It calculates the probability of machine failure based on sensor readings. If abnormal values are detected, the network predicts a high failure probability.
- 2) Imagine you are asked in an interview to design an image denoising solution. How would Markov Random Fields help model neighbourhood pixel dependencies?
 - A) Models the dependency between neighbouring pixels because nearby pixels usually have similar values. It identifies noisy pixels and estimates their correct values to produce a clean image.
- 3) If a company wants to build a speech recognition system. How would you apply a Hidden Markov Model to process sequential audio signals?
 - A) HMM represents speech as hidden states such as phonemes and uses audio features as observations. It finds the most likely sequence of sounds to recognize the spoken words.

4) During a technical interview, how would you explain the use of conditional Random Fields for sequence labeling tasks such as Part-of-Speech tagging?

A) CRF assigns labels like noun, verb, or adjective to words by considering neighbouring words and labels. This helps produce a more accurate and consistent POS-tagged sentence.